

DHRUV BAMAL

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PROFESSIONAL SUMMARY

Backend Engineer with systems design foundation and 200+ DSA problems solved in C++. Built production-grade APIs, databases, and distributed systems using Node.js, TypeScript, PostgreSQL, and Redis. Proficient in transaction safety, concurrent request handling, background job processing, and Docker deployment.

EDUCATION

B.Tech, CSE 2023 – 2027 • SRM Institute of Science and Technology, Ghaziabad • CGPA: 7.35/10

EXPERIENCE

Y Combinator Startup Simulation Forage Virtual Internship

Completed Y Combinator's working-startup simulation, learning early-stage dynamics, product-market fit, and entrepreneurial problem-solving.

SKILLS

Backend APIs: Node.js, NestJS, Express.js, REST API design, JWT & OAuth

Databases: PostgreSQL, SQL, Prisma ORM, Transaction safety, Row-level locking

Reliability: ACID transactions, Idempotency, Retry strategies, State machines

Background Jobs: Redis, BullMQ, Asynchronous queues

System Design: Scalability, Fault tolerance, Event-driven architecture

DevOps: Docker, Docker Compose, CI/CD (GitHub Actions)

Full Stack: React 18, Next.js, TypeScript

Languages: TypeScript, JavaScript, C++, Java, SQL, DSA

Tools: Git, GitHub, Jest, Supertest

PROJECTS

Multi-Tenant Approval Platform • Node.js, TypeScript, NestJS, PostgreSQL, Redis, BullMQ

- Built multi-tenant approval API with transaction-safe state machines, row-level locking, and idempotency keys.
- Designed PostgreSQL schema with role-based access control, audit trails, and per-tenant isolation; BullMQ workers handle escalations.

Appointment & Capacity-Reservation Backend • Node.js, TypeScript, NestJS, PostgreSQL, Redis, BullMQ

- Built atomic reservation algorithm: validate → lock capacity row → create reservation → decrement → all in one transaction.
- Prevented double-booking via ACID transactions and row-level locking; rate-limited endpoints with Redis.

Order-Import & Exception-Management Platform • Node.js, TypeScript, NestJS, PostgreSQL, BullMQ

- Designed async order-import API: CSV validation, batched processing, row-level error recording.
- Built duplicate-detection and retry with exponential backoff; exposed progress tracking, error reports, and audit logs.

ACHIEVEMENTS

- 1st Place, Prompt Rachna 2.0 — Techstacy 2.0 Hackathon, among 100+ participants (project: Helios Protocol)
- Solved 200+ problems on LeetCode, with consistent practice in C++